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Ethical Aspects of AI

Introduction to Artificial Intelligence

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Artificial Intelligence: a new hype







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12.09.2024

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http://t0.gstatic.com/images?q=tbn:ANd9GcRF2gcOUBAPRf0hfnlfm5LisSNnDZmPrgMqVvg5nwvl8ma4o2c3
http://www.insidesources.com/i-robot-whats-next/

Hype = Promotion by exaggerated claims

en.wikipedia.org/wiki/Hype



"Within a generation the problem of creating 'artificial intelligence' will be substantially solved."
(Marvin Minsky, 1967)

"Within 10 years computers won't even keep us as pets."
(Marvin Minsky, 1970)



"Artificial intelligence will reach human levels by around 2029. Follow that out further to, say, 2045, we will have multiplied the intelligence, the human biological machine intelligence of our civilization a billion-fold."
(Ray Kurzweil, 2005)

- Transhumanists / Posthumanists expect hyper-intelligent machines (singularity), cyborgs, immortality etc.

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https://pl.wikipedia.org/wiki/Marvin_Minsky
http://www.azquotes.com/author/10182-Marvin_Minsky
https://www.brainyquote.com/quotes/quotes/r/raykurzwei591137.html
https://en.wikipedia.org/wiki/Transhumanism
https://upload.wikimedia.org/wikipedia/commons/thumb/f/f4/Raymond_Kurzweil_Fantastic_Voyage.jpg/440px-Raymond_Kurzweil_Fantastic_Voyage.jpg

Reality:
AI applications are ubiquitous







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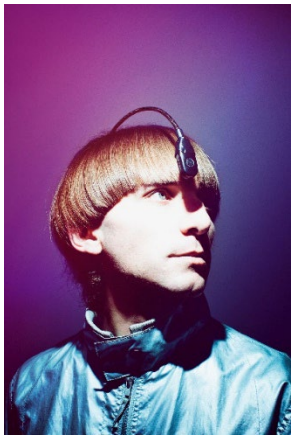
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https://en.wikipedia.org/wiki/Google_self-driving_car
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https://www.extremetech.com/wp-content/uploads/2017/09/Amazon-Echo-Blog_quotes_2-01.png

„Cyborg“: Technological enhancement of human body

Examples:

- Neil Harbisson: Antenna implanted in his skull allowing him to perceive colors that go beyond human perception



- Amanda Kitts wears an arm prosthesis



- Less futuristic examples: hearing aid, heart pacemaker, spectacles, etc.

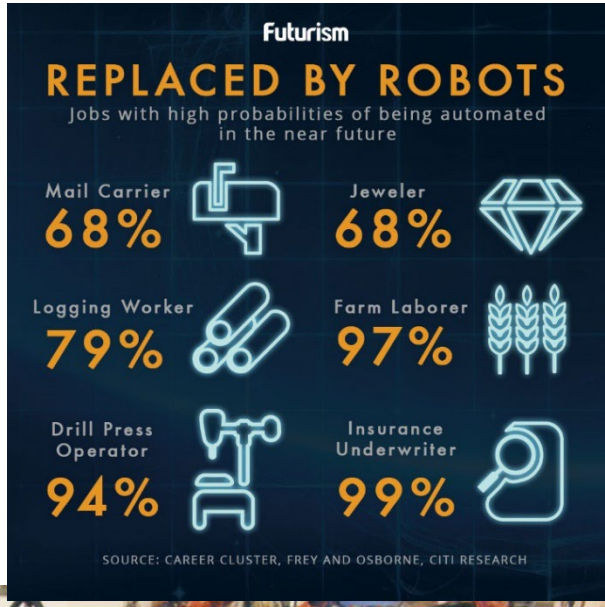
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https://en.wikipedia.org/wiki/Neil_Harbisson
https://en.wikipedia.org/wiki/Neil_Harbisson#/media/File:World's_First_Cyborg.jpg
http://4.bp.blogspot.com/-hJNV5BAgfh0/TqpzX-7lxQI/AAAAAAAALoc/vK0gRZlXihY/s1600/pr%25C3%25B3tesis+brazo+mano+bi%25C3%25B3nica+amanda+kitts.jpg

Machines replacing human labour: Impact on society?



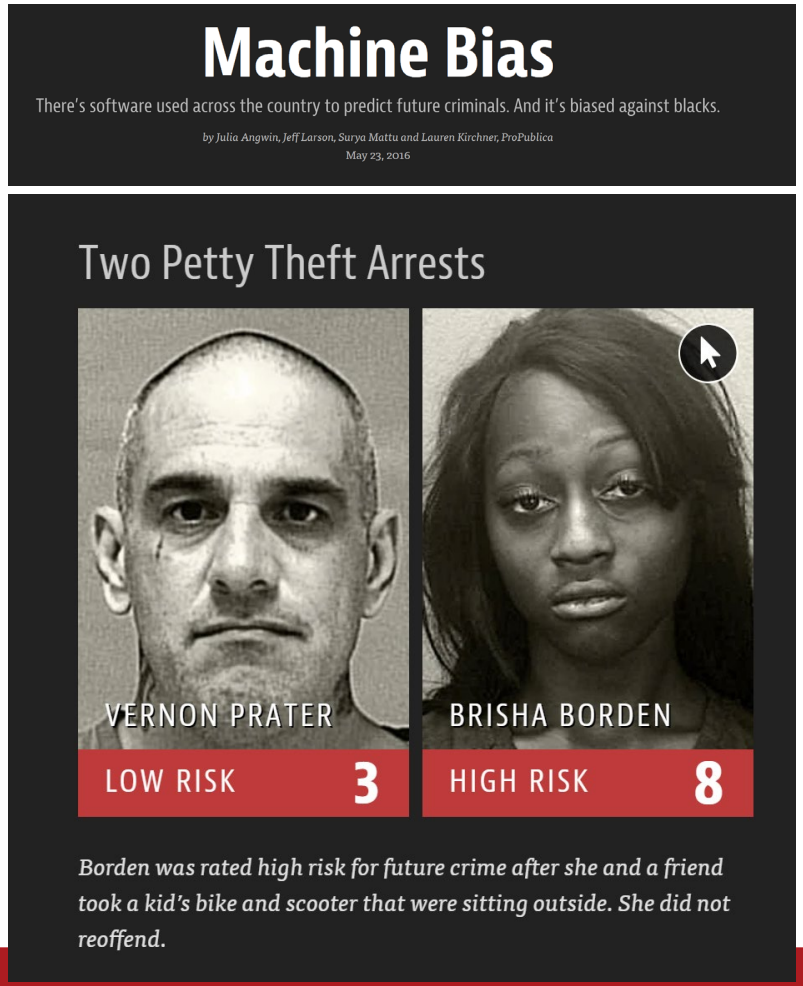




http://mastromaurowh.wikispaces.com/Industrial+Revolution+and+its+Impact
http://prodok40.de/
https://futurism.com/will-artificial-intelligence-take-job-experts-weigh/

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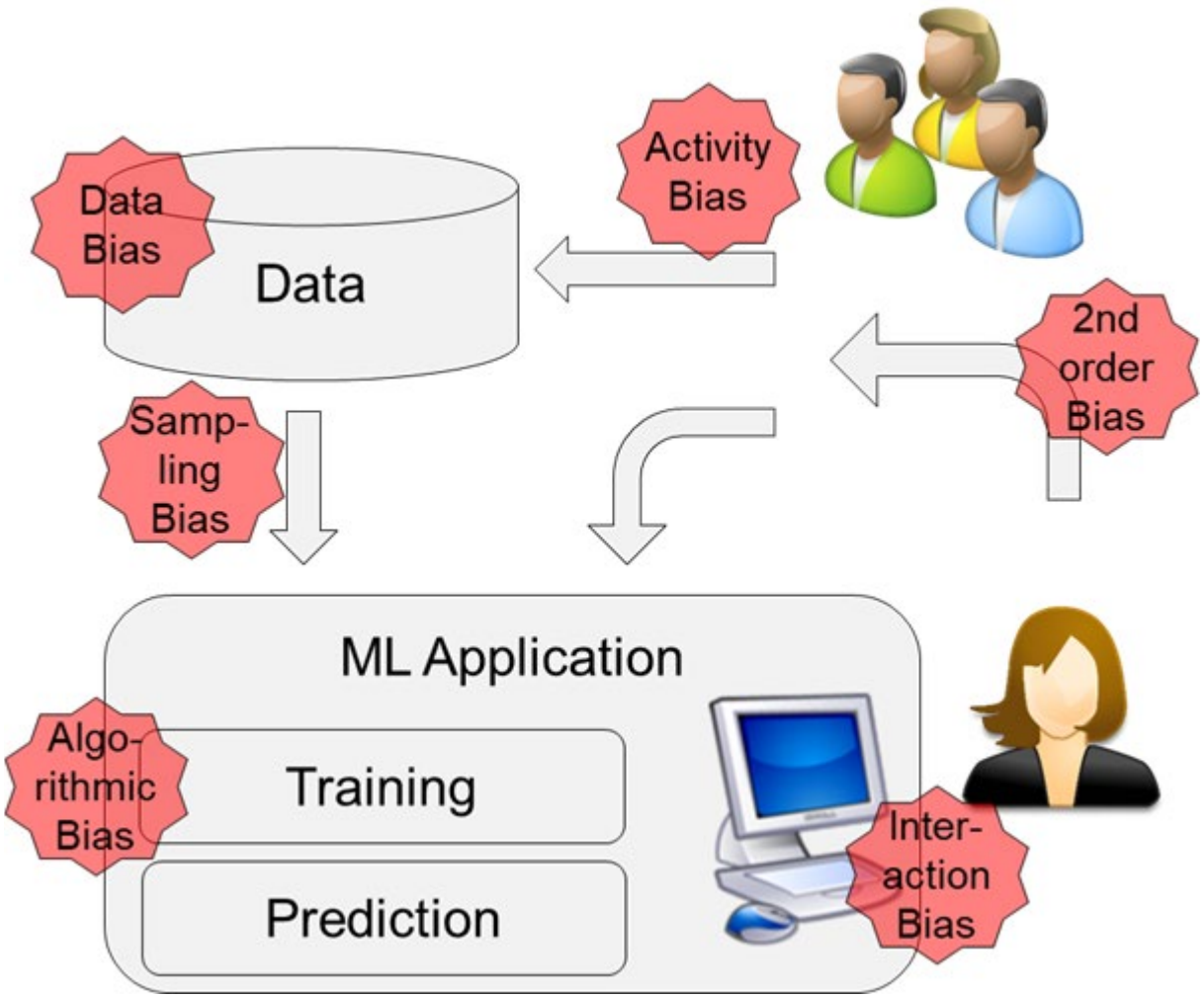
Biased machine learning: Projecting past prejudice to the future



https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing

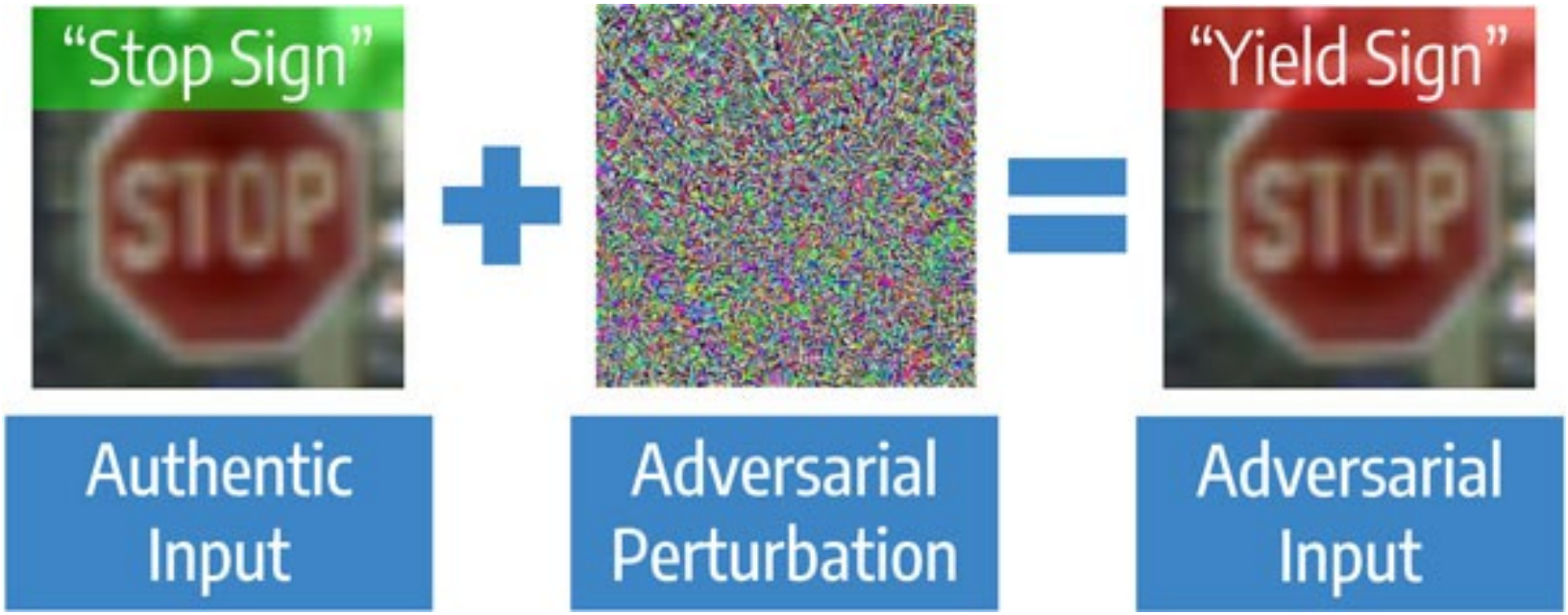
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Bias as vicious cycle



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Vulnerable AI: Attacking CV applications



https://ercim-news.ercim.eu/en116/special/detecting-adversarial-inputs-by-looking-in-the-black-box

Legal aspects: Who is to blame when a self-driving car crashes?



https://www.theaegisalliance.com/wp-content/uploads/Tesla-on-autopilot-crashes-into-police-car.jpg

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Green AI: Ever-increasing carbon footprint for training ML models

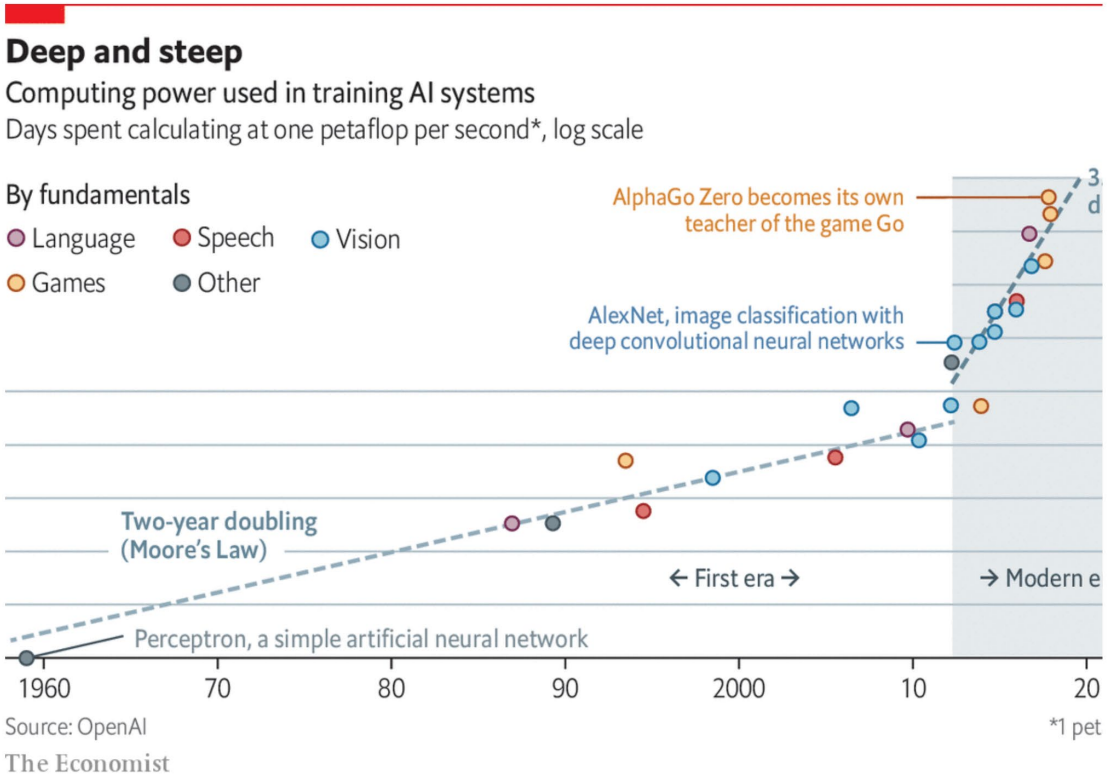
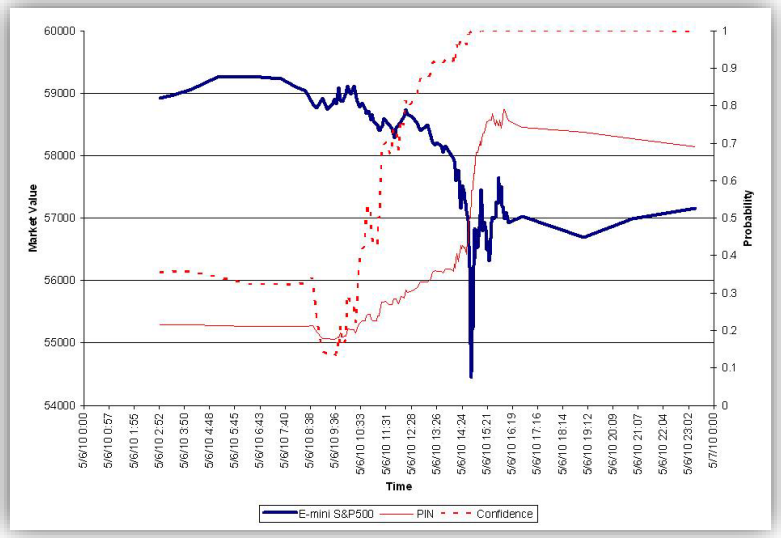


Fig. 3. Computing Power used in training AI Systems [Source: OpenAI.com [2], adapted in the Economist [3].]

Ethical question: Who needs to decide, human or machine?



High-frequency trading



Autonomous weapons



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